**MATHEMATICS — FINAL EXAM**
**MARKING SCHEME — teacher copy, not for distribution**
Grade 11AG · 26/10/2026 · Total: 100 marks · Anvarbek Khaydarov

Award method marks for a correct rule stated and applied, even where the final answer is wrong.

**SECTION A — Short answers (8 questions × 3 marks = 24 marks)**

1. Find $\lim_{x\to 2}(3x^2 - 5)$. [3 marks]
Answer: 7
2. Find $\lim_{x\to 3} (x^2 - 9)/(x - 3)$. [3 marks]
Answer: 6
3. Find $\lim_{x\to \infty} 1/x^2$. [3 marks]
Answer: 0
4. Find $\lim_{x\to \infty} (5x^2 + 1)/(2x^2 - x)$. [3 marks]
Answer: 5/2
5. Differentiate $y = 4x^5$. [3 marks]
Answer: $20x^4$
6. Differentiate $y = 3x^2 - 7x + 2$. [3 marks]
Answer: $6x - 7$
7. Differentiate $y = \sin x + e^x$. [3 marks]
Answer: $\cos x + e^x$
8. Differentiate $y = \ln x - \cos x$. [3 marks]
Answer: $1/x + \sin x$

**SECTION B — Show your working (8 questions × 5 marks = 40 marks)**

9. Find $\lim_{x\to 1} (x^2 + 2x - 3)/(x - 1)$. [5 marks]
Answer: 4
10. Find $\lim_{x\to 0} (\sqrt{(x + 4)} - 2)/x$. [5 marks]
Answer: 1/4
11. Find $\lim_{x\to \infty} (3x^3 - 2x)/(6x^3 + 5)$. [5 marks]
Answer: 1/2
12. Differentiate $y = (2x + 1)^5$. [5 marks]
Answer: $10(2x + 1)^4$
13. Differentiate $y = x^2e^x$, giving your answer in factorised form. [5 marks]
Answer: $xe^x(x + 2)$
14. Differentiate $y = (\sin x)/x$. [5 marks]
Answer: $(x \cos x - \sin x)/x^2$
15. Differentiate $y = \ln(3x^2 + 1)$. [5 marks]

Answer: $6x/(3x^2 + 1)$

16. Differentiate $y = \tan 4x$, and find the gradient of the curve at $x = 0$. [5 marks]

Answer: $4\sec^2 4x$; gradient 4

**SECTION C — Extended questions (4 questions × 9 marks = 36 marks)**

17. Evaluate the following limits:

(a) Find $\lim_{x \to 2} (x^3 - 8)/(x^2 - 4)$. [3 marks]

Answer: 3

(b) Find $\lim_{x \to \infty} (\sqrt{(x^2 + 3x)} - x)$. [3 marks]

Answer: 3/2

(c) Explain why $\lim_{x \to 0} |x|/x$ does not exist. [3 marks]

Answer: The one-sided limits are 1 and −1, so they disagree

18. Differentiate, simplifying your answer where possible:

(a) $y = \sqrt{(x^2 + 9)}$ [3 marks]

Answer: $x/\sqrt{(x^2 + 9)}$

(b) $y = e^{\sin x}$ [3 marks]

Answer: $\cos x \cdot e^{\sin x}$

(c) $y = \ln(\cos x)$ [3 marks]

Answer: $-\tan x$

19. A curve has equation $y = x^2 \sin x$.

(a) Find $dy/dx$. [3 marks]

Answer: $2x \sin x + x^2 \cos x$

(b) Find $d^2y/dx^2$. [4 marks]

Answer: $2 \sin x + 4x \cos x - x^2 \sin x$

(c) Find the value of $dy/dx$ at $x = 0$. [2 marks]

Answer: 0

20. A curve has equation $y = (\ln x)/x^2$, for $x > 0$.

(a) Find $dy/dx$. [4 marks]

Answer: $(1 - 2 \ln x)/x^3$

(b) Show that $dy/dx = 0$ when $x = \sqrt{e}$. [3 marks]

Answer: $1 - 2 \ln x = 0$, so $\ln x = ½$ and $x = \sqrt{e}$

(c) State, with a reason, the value of $\lim_{x \to \infty} (\ln x)/x^2$. [2 marks]

Answer: 0 — $x^2$ grows far faster than $\ln x$